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Metal ion transporter SLC39A14-mediated ferroptosis and glycosylation modulate the tumor immune microenvironment: pan-cancer multi-omics exploration of therapeutic potential

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Abstract

Background Ferroptosis, an iron-dependent form of regulated cell death driven by lipid peroxidation, has emerged as a pivotal mechanism in cancer progression and therapeutic resistance. Concurrently, glycosylation, as a key post-translational modification, plays a critical role in regulating cell signaling, immune evasion, and metastasis. Although both processes are independently implicated in tumor biology, the intersection between ferroptosis and glycosylation remains largely unexplored.

Methods We performed a comprehensive pan-cancer multi-omics analysis, integrating bulk transcriptomics, epigenomics, and single-cell RNA sequencing datasets. Ferroptosis and glycosylation-related genes were curated from The Molecular Signatures Database (MSigDB), leading to the identification of the metal ion transporter, SLC39A14 (solute carrier family 39 member 14), as a key intersecting gene. A ferroptosis-related gene signature was constructed using machine learning and Cox regression models, followed by survival analyses, immune microenvironment profiling, and a pathway enrichment analysis across The Cancer Genome Atlas (TCGA) cohort.

Results SLC39A14 was found to be significantly upregulated across multiple tumor types and associated with a poor prognosis, immune-stromal infiltration, and ferroptosis resistance. Among all cancer types analyzed, glioblastoma

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multiforme (GBM) and kidney renal cell carcinoma (KIRC) exhibited the significantly prognostic associations and the most pronounced differential expression of SLC39A14. Single-cell analysis revealed that SLC39A14 expression was enriched between stromal and immune populations, hypoxic perivascular niches, confirming its microenvironment-specific functions. These findings were corroborated by DNA methylation data showing promoter hypomethylation of SLC39A14 in tumors compared to normal tissues. Functionally, SLC39A14 was highly enriched in pathways related to angiogenesis, the epithelial-to-mesenchymal transition, cytokine signaling, and oxidative stress adaptation including vascular endothelial growth factor (VEGF) and cell metabolism-related signaling. A nomogram integrating SLC39A14 expression with clinical parameters improved overall survival predictions.

Conclusions In this study, we identified SLC39A14 as a dual regulator at the interface of ferroptosis and glycosylation, with significant impacts on tumor microenvironmental remodeling and therapeutic resistance. By leveraging multi-omics and single-cell transcriptomic data, we establish SLC39A14 as a promising prognostic biomarker and therapeutic target, particularly in brain and kidney cancers where ferroptosis modulation could offer novel clinical strategies.

Keywords SLC39A14, Ferroptosis, Glycosylation, Tumor microenvironment, Immune infiltration, Pan-cancer analysis, Prognostic biomarker

Introduction

Ferroptosis is a unique form of regulated cell death driven by iron-dependent lipid peroxidation, setting it apart from traditional forms of cell death such as apoptosis and necrosis [1]. This process is characterized by the accumulation of reactive oxygen species (ROS) and the depletion of antioxidant systems, particularly glutathione, leading to catastrophic oxidative damage. In cancer, ferroptosis plays a dual role, acting as a tumor suppressor in some contexts while promoting therapy resistance and immune evasion in others. Separately, glycosylation is a critical post-translational modification that governs protein folding, stability, and signaling, and immune recognition. It involves the enzymatic addition of sugar moieties (glycans) to proteins and lipids, primarily occurring in the endoplasmic reticulum (ER) and Golgi apparatus. Glycosylation is broadly classified into N-linked and O-linked forms, depending on the site of sugar attachment, and it regulates a wide array of biological processes, including cell growth and cell communication, receptor activation, immune recognition, and extracellular matrix (ECM) organization. Aberrant glycosylation is a hallmark of cancer, frequently contributing to tumor progression, the epithelial-to-mesenchymal transition (EMT), and immune escape [2]. For instance, altered glycosylation patterns of immune checkpoint molecules such as programmed death ligand 1 (PD-L1) and cytotoxic T-lymphocyte-associated protein (CTLA-4) can impair T-cell activation and facilitate immune escape. Furthermore, increased sialylation and fucosylation in tumor cells have been linked to greater invasiveness and resistance to therapy.

Recent studies highlighted a potential interplay between glycosylation and ferroptosis regulation, suggesting that post-translational modifications may influence ferroptotic susceptibility and cellular redox

homeostasis. Glycosylation of key proteins, such as membrane transporters, antioxidant enzymes, and lipid metabolism regulators, can modulate the redox balance, iron uptake, and membrane integrity, all of which are central determinants of ferroptotic vulnerability. Notably, manganese, an essential cofactor for several glycosyltransferases, has emerged as a critical regulator of both glycosylation processes and oxidative stress responses. This biochemical intersection implies that glycosylation may modulate ferroptosis either directly through enzymatic machinery or indirectly via microenvironmental remodeling and immune evasion, although the exact mechanisms remain underexplored [3–5].

By curating gene sets from the Molecular Signatures Database (MSigDB) and intersecting ferroptosis and glycosylation-related genes, we identified SLC39A14 as the major overlapping gene, suggesting a unique dual role in controlling the redox balance and post-translational modifications. SLC39A14 is a membrane transporter that facilitates the uptake of non-transferrin-bound iron (NTBI) and manganese, in addition to its primary role in zinc transport. Through its regulation of metal ion homeostasis, SLC39A14 can influence both ferroptosis susceptibility via iron accumulation and ROS generation, and glycosylation processes by modulating manganese availability for glycosyl transferases. In this study, we aimed to better understand the biological and clinical relevance of SLC39A14 by exploring its connection to ferroptosis-related gene signatures, its relationship with the tumor immune and stromal microenvironment, and its role in promoting ferroptosis resistance and immune-stromal remodeling. To this end, we performed a comprehensive pan-cancer analysis using transcriptomic data from public database, complemented by single-cell transcriptomic profiling (Fig. 1).

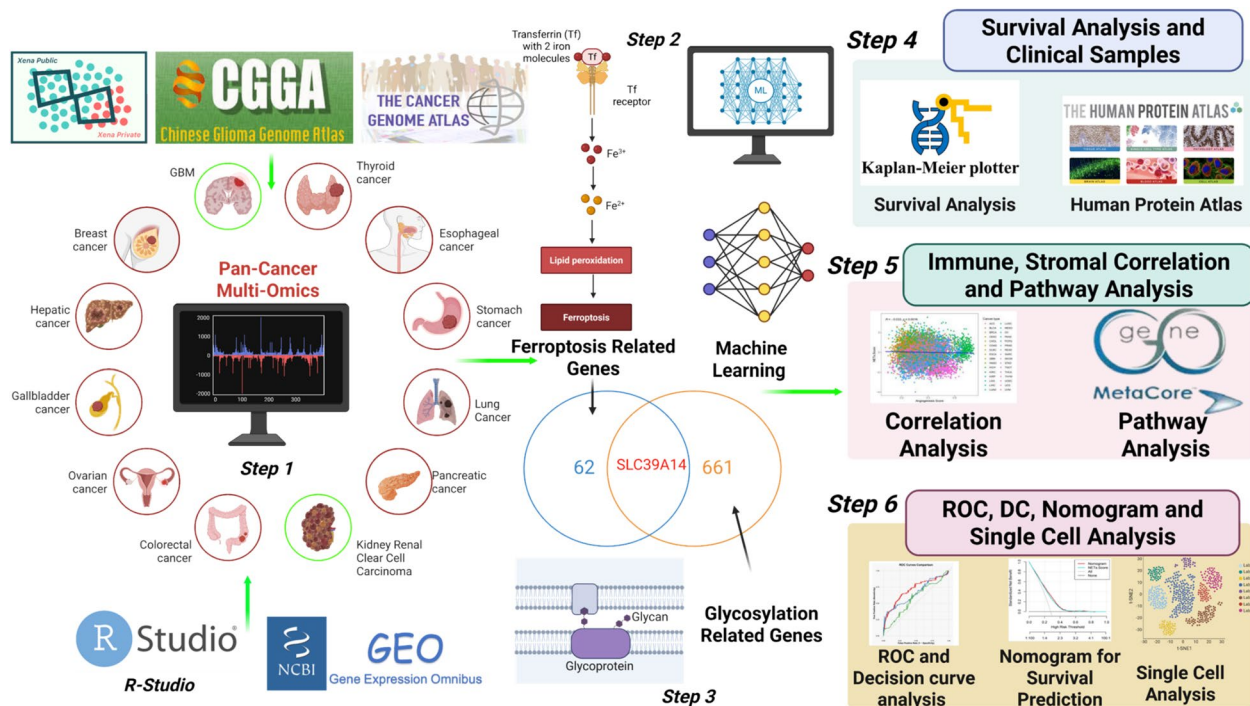


Fig. 1 Schematic of the integrated bioinformatics workflow for identifying key genes linking glycosylation and ferroptosis in a pan-cancer analysis. RNA-sequencing data from TCGA and UCSC Xena databases were analyzed to identify candidate genes from two distinct biological processes: glycosylation (663 genes) and ferroptosis (64 genes). A subsequent analysis integrated differential gene expression profiling, survival curves, immune and stromal correlations, and pathway enrichment. Downstream steps included nomogram construction for survival predictions, ROC curve predictions, and single cell analyses. This workflow highlights the prognostic and therapeutic potential of SLC39A14 as a critical mediator bridging glycosylation and ferroptosis in cancer progression

Materials and methods

Gene expression data and survival analysis

Normalized RNA-sequencing (RNA-Seq) data and clinical annotations for 33 cancer types ($n > 10,000$) were downloaded from TCGA via UCSC Xena (<http://xena.ucsc.edu>) [6]. Expression data were $\log_2$ -transformed as $\log_2(\text{TPM} + 1)$ for normalization [7]. Clinical endpoints included overall survival (OS), disease-specific survival (DSS), and the progression-free interval (PFI). Differential expression of SLC39A14 between tumor and normal tissues was analyzed, alongside co-expression analyses of ferroptosis-related genes. Kaplan–Meier and univariate Cox regression analyses were used to assess the prognostic value of SLC39A14, with patients stratified by median expression levels. A prognostic nomogram incorporating SLC39A14, ferroptosis scores, and clinical variables were built using multivariate Cox models. The performance was evaluated via calibration curves, receiver operating characteristic (ROC) curves, and a decision curve analysis (DCA), with cross-validation applied to ensure model robustness [8–10].

Data collection and preprocessing

To investigate the role of glycosylation and ferroptosis-related genes in cancer, glycosylation-related genes

($n = 663$) and ferroptosis-related genes ($n = 64$) were retrieved from MSigDB (<https://www.gsea-msigdb.org/gsea/msigdb/human/>) respectively using the keywords "glycosylation" and "ferroptosis" [11]. After removing duplicates, the two gene sets were intersected using R (v4.1.2) to identify overlapping candidates. We identified as common to both pathways and were selected for further analysis, and their relevance to cancer was supported through a literature review and database validation.

Protein expression and DNA methylation analysis

Protein-level expression of SLC39A14 across various cancer types was examined using immunohistochemistry (IHC) and immunofluorescence (IF) data from the Human Protein Atlas (HPA). The HPA integrates antibody-based imaging, transcriptomic, and proteomic data to provide comprehensive insights into protein localization and expression in human tissues and cell types [12]. The DNA methylation data for SLC39A14 cytosine-phosphate-guanine (CpG) sites were obtained from TCGA and analyzed using the MethSurv database (<https://biit.cs.ut.ee/methsurv/>) to evaluate associations with gene expressions and clinical outcomes [13]. A differential methylation analysis was performed to explore the epigenetic regulation of SLC39A14, with the aim of

uncovering mechanisms underlying its altered expression in tumors compared to normal tissues.

Immune infiltration and functional enrichment analyses

The ESTIMATE (Estimation of STromal and Immune cells in MAlignant Tumors using Expression data) algorithm was applied to quantify stromal and immune cell infiltration across pan-cancer datasets [14]. A Pearson correlation analysis was used to assess the association between SLC39A14 expression and immune/stromal scores, using the TIMER web platform (<https://cistrome.shinyapps.io/timer/>) [15]. Scatterplots with regression lines were generated to visualize correlation trends [16], with a focus on selected cancer types including GBM (glioblastoma multiforme), kidney renal cell carcinoma (KIRC), and HCC (hepatocellular carcinoma). To explore biological functions associated with SLC39A14, differentially expressed genes (DEGs) were identified using the "limma" R package, based on high vs. low SLC39A14 expressions. Functional enrichment analyses were then conducted using the "clusterProfiler" package. Gene ontology (GO) enrichment included biological processes (BPs), molecular functions (MFs), and cellular components (CCs), while a Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis identified relevant signaling pathways [17–22]. Terms with an adjusted false discovery rate (FDR) of <0.05 were considered statistically significant. Additionally, MetaCore (<https://portal.genego.com>) was used to investigate pathway maps, biomarker networks, and cell signaling processes related to SLC39A14, as previously described [23–25].

Single-cell RNA-seq analysis

Single-cell transcriptomic analyses were performed using two publicly available datasets: GSE159115 (KIRC) [26] and GSE182109 (GBM) [27]. Pre-filtered, processed, and annotated expression matrices were respectively obtained from CellxGene and the Broad Institute Single Cell Portal. The GSE159115 dataset, originally in.h5ad format, was converted to.rds using the sceasy R package. The GSE182109 dataset was processed directly using the standard Seurat pipeline [28–30]. Dimensionality reduction was conducted using both t-distributed stochastic neighbor embedding (t-SNE) and uniform manifold approximation and projection (UMAP), based on the top 20 principal components (PCs) to capture major transcriptional variability. Original cell-type annotations were preserved by merging downloaded metadata with the constructed Seurat object. All downstream analyses, including clustering, visualization, and interpretation, were performed using the Single Cell Pipeline (SCP) R package [31–34].

Statistical analysis

All bioinformatics analyses were conducted using a combination of publicly available online platforms and locally installed software. Data visualization and statistical analyses were performed using ggplot2 package in R Studio [35], SPSS (IBM, Armonk, NY, USA) [36], and ImageJ software (NIH, Bethesda, MD, USA). Additional analyses were supported by Omics Playground v3.4.1 and SRPlot. Quantitative data are presented as the mean $\pm$ standard deviation (SD), based on a minimum of three independent experiments [37–39]. One-way and two-way analyses of variance (ANOVAs) were used to assess differences between groups, with the Bonferroni correction applied for multiple comparisons. Survival analyses were conducted using the Kaplan–Meier method, and statistical significance was determined using the log-rank test. A p value of <0.05 was considered statistically significant unless otherwise indicated, as described in previous studies [40–42].

Results

Construction and validation of a prognostic ferroptosis-related gene signature

To systematically assess the clinical relevance of ferroptosis in cancer, we curated a list of 50 genes previously reported to participate in ferroptosis regulation. Using LASSO Cox regression modeling on TCGA pan-cancer datasets, we identified an optimized 15-gene prognostic signature for survival predictions. This approach applied penalization to reduce overfitting while maximizing the predictive power. The final panel included two ferroptosis suppressors, reflecting the dual roles of ferroptosis regulation in tumor biology (Supplementary Figure S1A). The optimal regularization parameter ($\lambda = 0.0014$) was selected via tenfold cross-validation to minimize partial likelihood deviance (Supplementary Figure S1B). A ferroptosis score was then computed for each patient as a weighted sum of gene expression levels, with weights derived from the model's coefficients. A pairwise Spearman correlation analysis revealed both positive and negative interactions among the 15 genes, indicating complex co-regulation. Individual gene contributions to survival were further evaluated using a univariate Cox regression analysis (Supplementary Figure S1C). Genes exhibited hazard ratios (HRs) exceeding 1.2, indicating that their higher expression levels were associated with worse survival. In contrast, other genes demonstrated HRs of <1 , suggesting protective effects potentially mediated through promoting ferroptosis or tumor-suppressive functions in specific cancer contexts. These variations in prognostic impacts underscore the functional heterogeneity within the ferroptosis gene network and highlight that not all ferroptosis-related genes act uniformly across cancer types: some facilitate tumor growth, while others

appear to counteract it. These findings suggest that the model captures a biologically relevant spectrum of ferroptosis activity with prognostic implications. Interestingly, for certain genes within the signature, the observed prognostic associations may diverge from cancer type-specific reports in previous studies. This phenomenon could be partly attributable to the pan-cancer design of our analysis, where the same ferroptosis-related gene can exert distinct biological effects depending on the tumor context. In one cancer type, elevated expression may facilitate ferroptosis and exert tumor-suppressive effects, whereas in another type, it may contribute to tumor progression through altered metabolic dependencies or microenvironmental adaptations. When aggregated across multiple heterogeneous cancer types, these context-dependent effects can be diluted or, in some cases, reversed in the overall statistical output. This highlights the necessity of further exploration using analyses stratified by tumor type.

Integrative pan-cancer analysis

To investigate the heterogeneity and clinical relevance of ferroptosis across different cancer types, we analyzed ferroptosis-related scores in 32 TCGA cancers. As shown in the top panel of Fig. 2A, the distribution of the NETs score varied considerably among cancers, with relatively high median values in Uterine carcinosarcoma (UCS), GBM, Head and neck squamous cell carcinoma (HNSC), and Lung squamous cell carcinoma (LUSC), whereas Prostate adenocarcinoma (PRAD) and Thyroid carcinoma (THCA) exhibited the lowest values, with many patients displaying negative scores. To systematically evaluate prognostic relevance, we performed pan-cancer survival analyses. Bubble plots demonstrated that the risk score was significantly associated with PFI, DSS, OS in multiple cancer types, with the associations observed in KIRC and GBM. Cox regression analysis further confirmed these findings. In the analysis of PFI, high scores predicted shorter time to recurrence or progression in multiple cancers (Fig. 2B). Similar trends were observed for DSS (Fig. 2C), and OS (Fig. 2D). Kaplan–Meier survival analyses in both the training (Fig. 2E) and validation (Fig. 2F) cohorts from TCGA showed that patients with high ferroptosis scores experienced significantly worse survival outcomes compared to those with lower scores ($p < 0.001$). Collectively, these results indicate that the ferroptosis-related risk score is broadly applicable across cancer types but exerts especially strong prognostic impact in KIRC, and GBM thereby supporting the ferroptosis score as a robust pan-cancer biomarker for patient risk stratification.

Cancer type-specific prognostic implications

We next examined cancer type-specific prognostic implications of the ferroptosis score in more detail. In GBM, patients with higher scores had shorter survival intervals, suggesting that ferroptosis inhibition contributes to the aggressive phenotype of these tumors (Fig. 3A). In KIRC, all survival endpoints showed p values of < 0.001 , underlining the impacts of iron metabolism and lipid peroxidation suppression in renal tumor biology (Fig. 3B). In LGGs, high ferroptosis scores were linked to poor survival outcomes across all endpoints, with p values of < 0.001 (Fig. 3C), and KIRP in Fig. 3D. These results are particularly important given the known metabolic plasticity and redox sensitivity of glioma cells. These patterns across multiple cancer types suggest that ferroptosis regulation may represent a common driver of tumor aggressiveness and could serve as a useful indicator of therapeutic vulnerability.

Development and validation of a prognostic nomogram

To enhance clinical applicability, we developed a nomogram that integrates ferroptosis scores with key clinical parameters including patient age, tumor stage, and cancer type. The nomogram assigns weighted points to each variable and calculates a total risk score, which can then be used to estimate 1-, 3-, 5-, and 10-year survival probabilities. The nomogram structure shows the cumulative contribution of each factor (Fig. 4A). The ferroptosis score emerged as the most influential variable, reaffirming its biological and prognostic significance. Calibration curves (Fig. 4B) showed excellent agreement between predicted and observed survival rates across multiple time points. The ROC curve analysis (Fig. 4C) revealed high predictive accuracy, with area under the ROC curves (AUROCs) of 0.78 in the training cohort and 0.81 in the test cohort at the 5-year mark. Time-dependent AUROC analyses (Fig. 4D) further demonstrated that the nomogram outperformed the ferroptosis score alone across all time intervals. A decision curve analysis (DCA) (Fig. 4E, F) confirmed the model's superior net clinical benefit, especially in intermediate-to-high risk ranges, where clinical decision-making is often uncertain. These results suggest that integrating molecular signatures, such as ferroptosis scores, with standard clinical features can improve survival predictions and inform individualized patient management.

Correlations between ferroptosis scores and malignant features

To elucidate the biological implications of ferroptosis scores, we assessed their correlations with hallmark cancer traits across pan-cancer data. Significant positive correlations were found between ferroptosis scores and pathways related to the EMT, angiogenesis, and cell-cycle

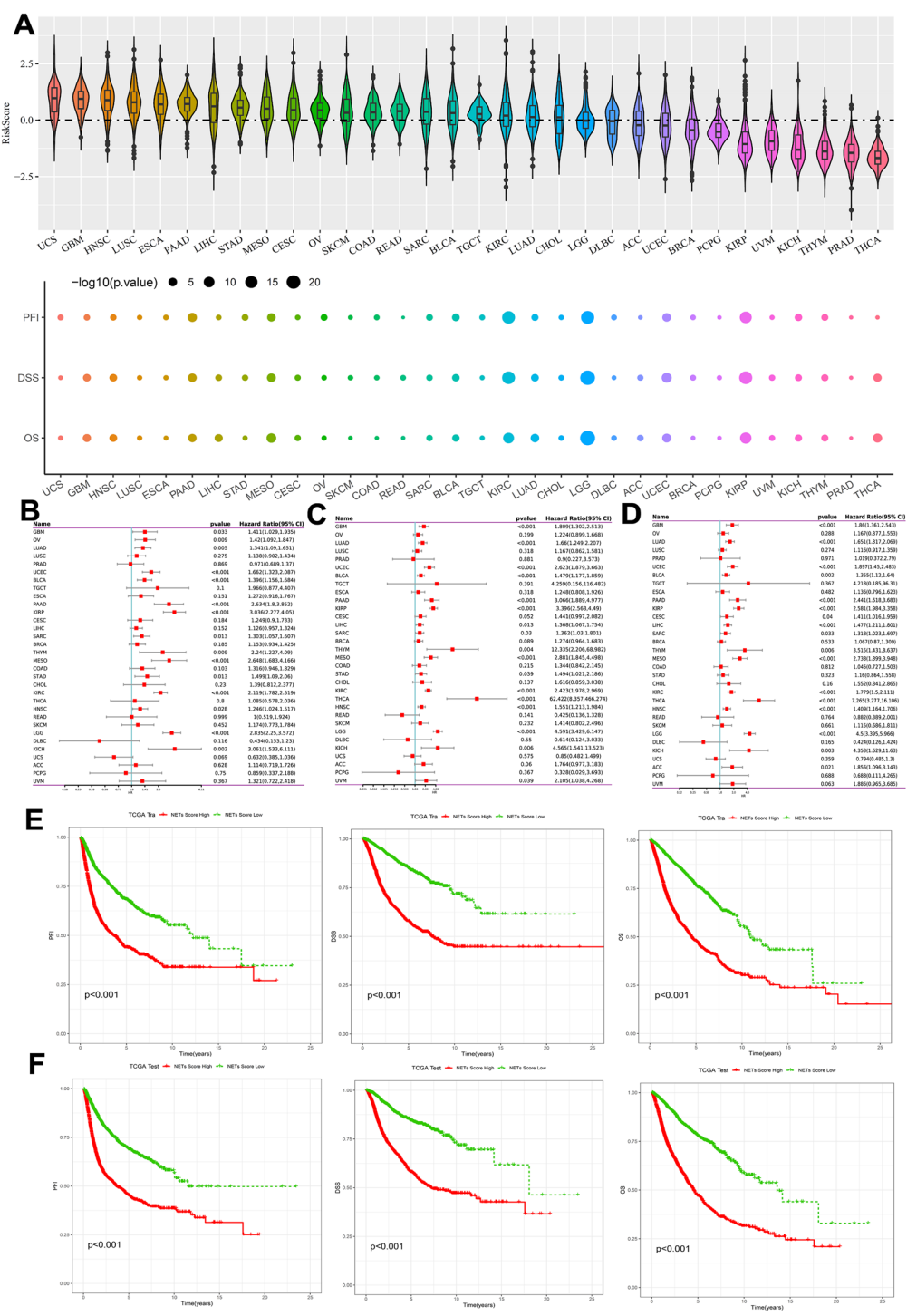


Fig. 2 Pan-cancer evaluation of the ferroptosis-related prognostic signature (risk score) across different cancer types. **(A)** Violin plots (upper panel) showing the distribution of risk scores across TCGA cohorts. The lower panel presents bubble plots illustrating correlations between risk scores and clinical outcomes, including progression-free interval (PFI), disease-specific survival (DSS), and overall survival (OS). Bubble sizes reflect the significance level ($-\log_{10}$ p value). **(B–D)** Forest plots summarizing univariate Cox regression analyses for risk scores across cancer types, presenting hazard ratios (HRs) and 95% confidence intervals (CIs). Red dots represent increased risk (HR > 1), whereas green dots indicate significantly reduced risk (HR < 1) **(B)** PFI, **(C)** DSS, **(D)** OS. **(E–F)** Kaplan–Meier survival curves comparing PFI, DSS, and OS between high (red) and low (green) ferroptosis score groups in representative cancers of **(E)** training and **(F)** validation cohorts, highlighting the prognostic relevance of the ferroptosis signature

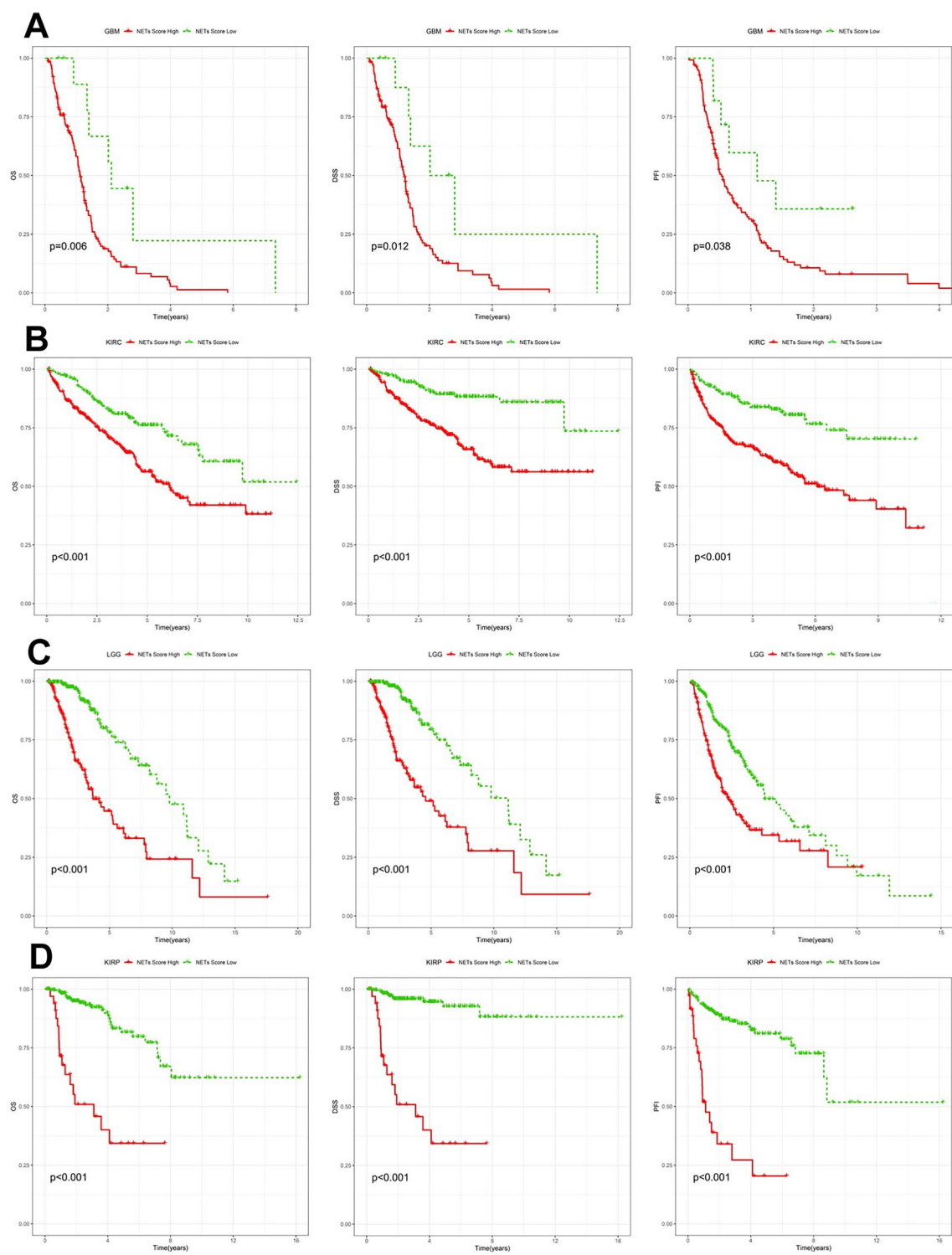


Fig. 3 Kaplan–Meier survival analysis of the ferroptosis-related prognostic signature (ferroptosis score) in four representative cancers. Kaplan–Meier curves comparing high (red) and low (green) ferroptosis score groups for overall survival (OS; left panels), disease-specific survival (DSS; middle panels), and the progression-free interval (PFI; right panels) in **(A)** GBM, **(B)** KIRC, **(C)** LGGs and **(D)** KIRP. In all four cancer types, higher ferroptosis scores were significantly associated with worse prognoses ($p < 0.05$, log-rank test), indicating the prognostic value of the ferroptosis-related gene signature across multiple survival endpoints

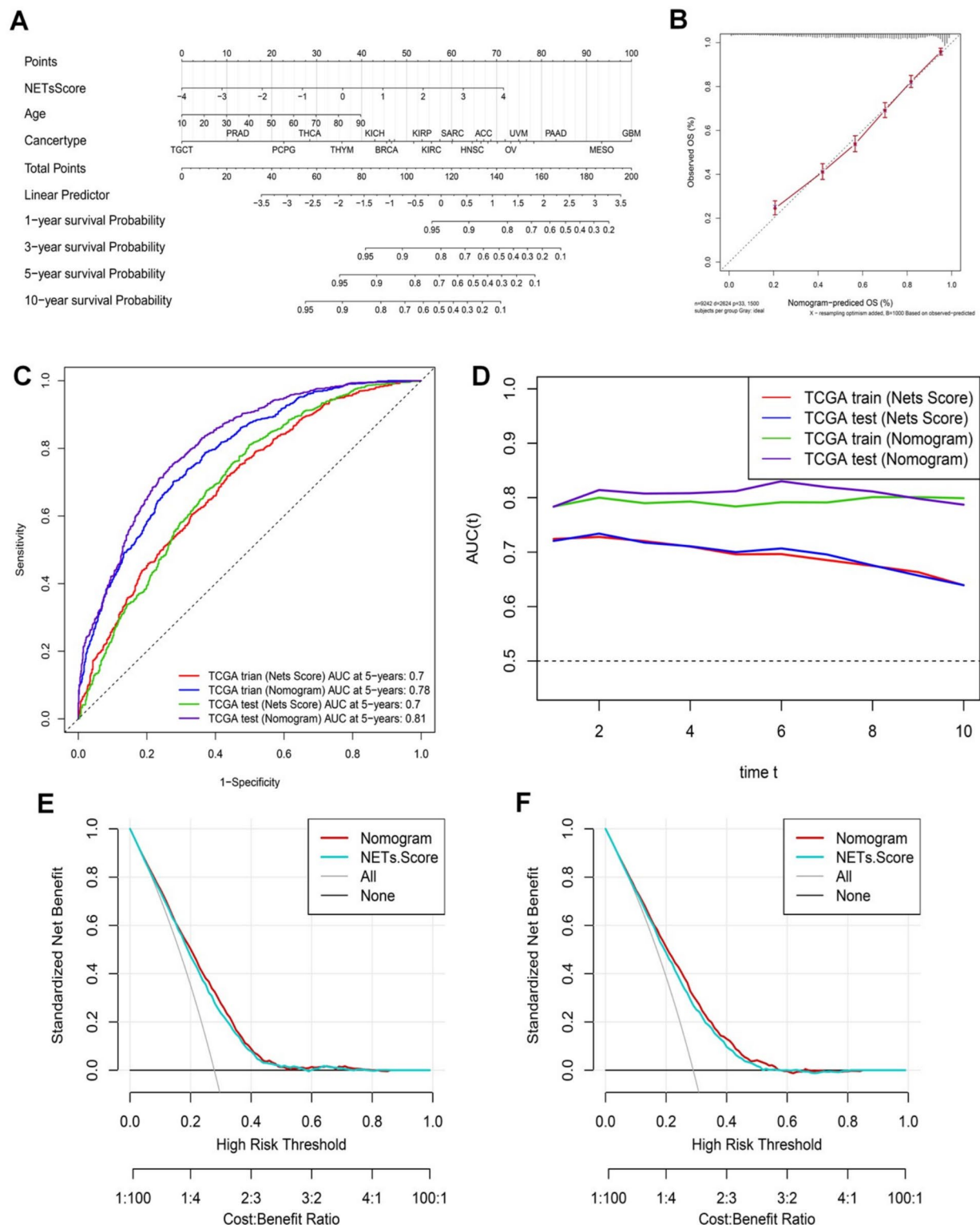


Fig. 4 Development and validation of a ferroptosis-based nomogram for multi-cancer survival predictions. **(A)** Nomogram integrating the ferroptosis NET score, patient age, and cancer type to estimate 1-, 3-, 5-, and 10-year OS probabilities. **(B)** Calibration curve comparing nomogram-predicted OS probabilities with observed outcomes; the dashed diagonal line denotes a perfect prediction. **(C)** Receiver operating characteristic (ROC) curves for 5-year OS predictions in TCGA training and test cohorts, comparing nomogram and NET score models. **(D)** Time-dependent ROC curves showing predictive accuracy over 10 years for the nomogram and NET scores in TCGA training and test sets. Decision curve analysis (DCA) illustrating the net clinical benefit of using the nomogram versus NET scores alone across different threshold probabilities in TCGA training **(E)** and test **(F)** cohorts. The nomogram consistently demonstrated superior predictive utility, underscoring its potential for individualized prognosis assessment

progression. As shown in Fig. 5A–C, ferroptosis scores were moderately associated with these processes across various tumor types such as the EMT, angiogenesis, and cell cycle. These processes are known to promote tumor invasiveness, immune escape, and resistance to treatment. Cancer-specific analyses (Fig. 5D–F) demonstrated correlations with the EMT and angiogenesis in brain and kidney cancers showed links to cell-cycle dysregulation. These observations support the hypothesis that suppression of ferroptosis enhances malignant phenotypes, possibly by stabilizing tumor cell membranes, modulating redox-sensitive transcription factors such as NRF2, or promoting immunosuppressive microenvironments.

Identification of SLC39A14 as a key gene in ferroptosis and glycosylation

By intersecting glycosylation- and ferroptosis-related gene sets from MSigDB, we identified two overlapping genes. SLC39A14 was prioritized due to its stronger expression across multiple tumor types and its established role in metal ion transport, particularly zinc and manganese, which are critical in redox regulation. SLC39A14 was overexpressed in tumors such as GBM and KIRC relative to adjacent normal tissues (Fig. 6A, B). The univariate Cox analysis revealed that elevated SLC39A14 expression was significantly associated with poor OS and DSS. These findings suggest that SLC39A14 may suppress ferroptosis through mechanisms related to intracellular metal ion homeostasis and membrane stability, thereby promoting tumor progression and poor clinical outcomes (Fig. 6C, D).

DNA methylation and protein-level validation of SLC39A14 expression

To elucidate the regulatory mechanisms of SLC39A14 expression, we analyzed DNA methylation data from TCGA. Heatmap visualization (Fig. 7A, B) revealed significant hypomethylation of multiple CpG sites within the promoter region of SLC39A14 in tumor tissues compared to adjacent normal controls. This hypomethylation pattern exhibited inverse correlations with gene expression levels, suggesting that epigenetic silence is alleviated in renal tumors, thereby leading to SLC39A14 upregulation. Stratified analyses based on clinical parameters, including age, sex, and ethnicity, confirmed that promoter hypomethylation remained a robust predictor of gene expression independent of demographic factors. These findings indicate that SLC39A14 overexpression in renal tumors is primarily driven by epigenetic reprogramming rather than patient-specific characteristics. To validate the translational relevance of these findings, we examined IHC staining images (Fig. 7C, D) from the HPA which demonstrated SLC39A14 protein expression in clinical tissue samples. Compared to normal kidney

tissues, kidney cancer samples exhibited SLC39A14 staining and increased positivity, indicating elevated SLC39A14 expression in cancerous tissues. Interestingly, glioma samples showed minimal staining, suggesting a potential tumor-type-specific role of SLC39A14 in oncogenesis. To further investigate the subcellular localization of SLC39A14 in a pan-cancer analysis, IF staining was performed in multiple cancer cell lines using the HPA database. However, due to resource limit, the IF image including A-431 (human epidermoid carcinoma), U-251 MG (human glioblastoma), and U-2 OS (human osteosarcoma) cells (Fig. 7E). A confocal microscopic analysis demonstrated that SLC39A14 was predominantly localized to the cytoplasm, with co-localization observed in microtubules and the ER. This subcellular distribution aligns with the established role of SLC39A14 as a metal ion transporter implicated in maintaining redox homeostasis and regulating ferroptosis. The cytoplasmic localization further suggests an active role in intracellular metal ion trafficking, which is critical for cellular metabolism and stress response mechanisms. These findings suggest that promoter hypomethylation of SLC39A14 contributes to its upregulation in renal tumors, highlighting the role of epigenetic reprogramming in its dysregulation. The increased protein expression in renal tumor tissues, along with its cytoplasmic localization, underscores its potential involvement in metal ion transport and redox balance processes that are fundamental for ferroptosis regulation and cancer cell survival. These observations establish mechanistic links between epigenetic modifications and SLC39A14 expression, and its functional significance in tumor pathophysiology.

Functional enrichment and SLC39A14 expression with stromal components

We conducted a functional enrichment analysis on DEGs between high- and low-ferroptosis scoring groups. The high-scoring group was significantly enriched in GO biological processes such as nuclear division, chromosome segregation, and cytokine and receptor interactions. A KEGG pathway analysis highlighted the interleukin (IL)–17 signaling pathway, cell cycle signaling, suggesting that ferroptosis-suppressed tumors may upregulate inflammatory and proliferative pathways to drive tumor growth and immune modulation (Supplementary Figure S2A). In contrast, the low-scoring group exhibited enrichment in hormonal signaling, dopaminergic synapse, and insulin secretion (Supplementary Figure S2B). GO terms such as male sex differentiation and genitalia development suggested that these tumors retain more-differentiated and less-proliferative characteristics. These results highlight that tumors with high ferroptosis scores are more proliferative, immune-reactive, and potentially

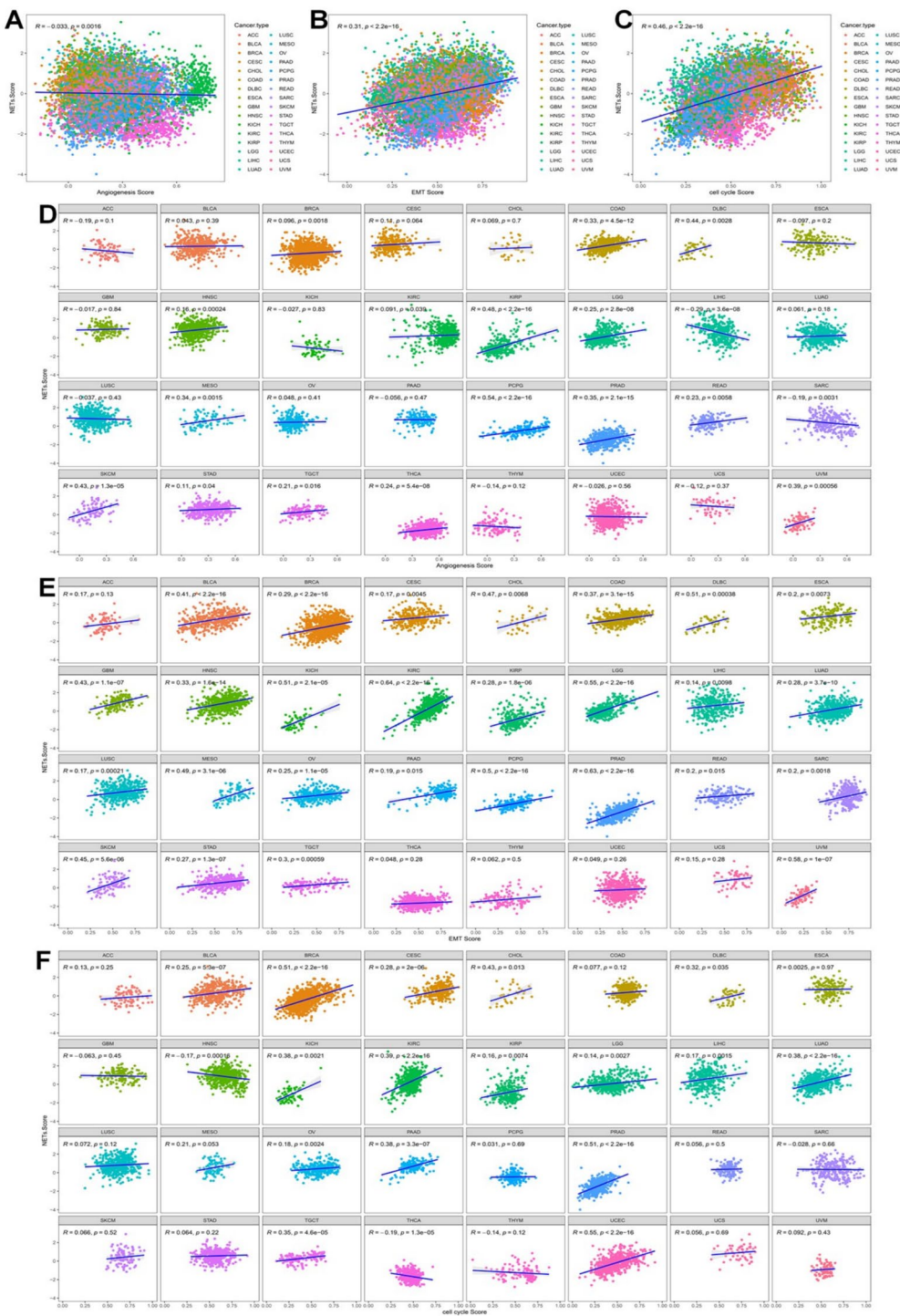


Fig. 5 Pan-cancer correlation of ferroptosis scores (NET scores) with malignant cancer hallmarks. **(A–C)** Pan-cancer scatterplots showing associations between ferroptosis scores and hallmark biological processes: **(A)** angiogenesis, **(B)** epithelial-to-mesenchymal transition (EMT), and **(C)** cell cycle. Each plot presents Pearson's correlation coefficients (R) and corresponding p values, revealing significant positive correlations in multiple cancer types. Detailed cancer type-specific scatterplots assessing correlations between ferroptosis scores and **(D)** angiogenesis, **(E)** EMT, and **(F)** cell-cycle signatures. Positive associations across various tumors suggest that elevated ferroptosis scores are linked to enhanced malignant phenotypes and may contribute to tumor progression

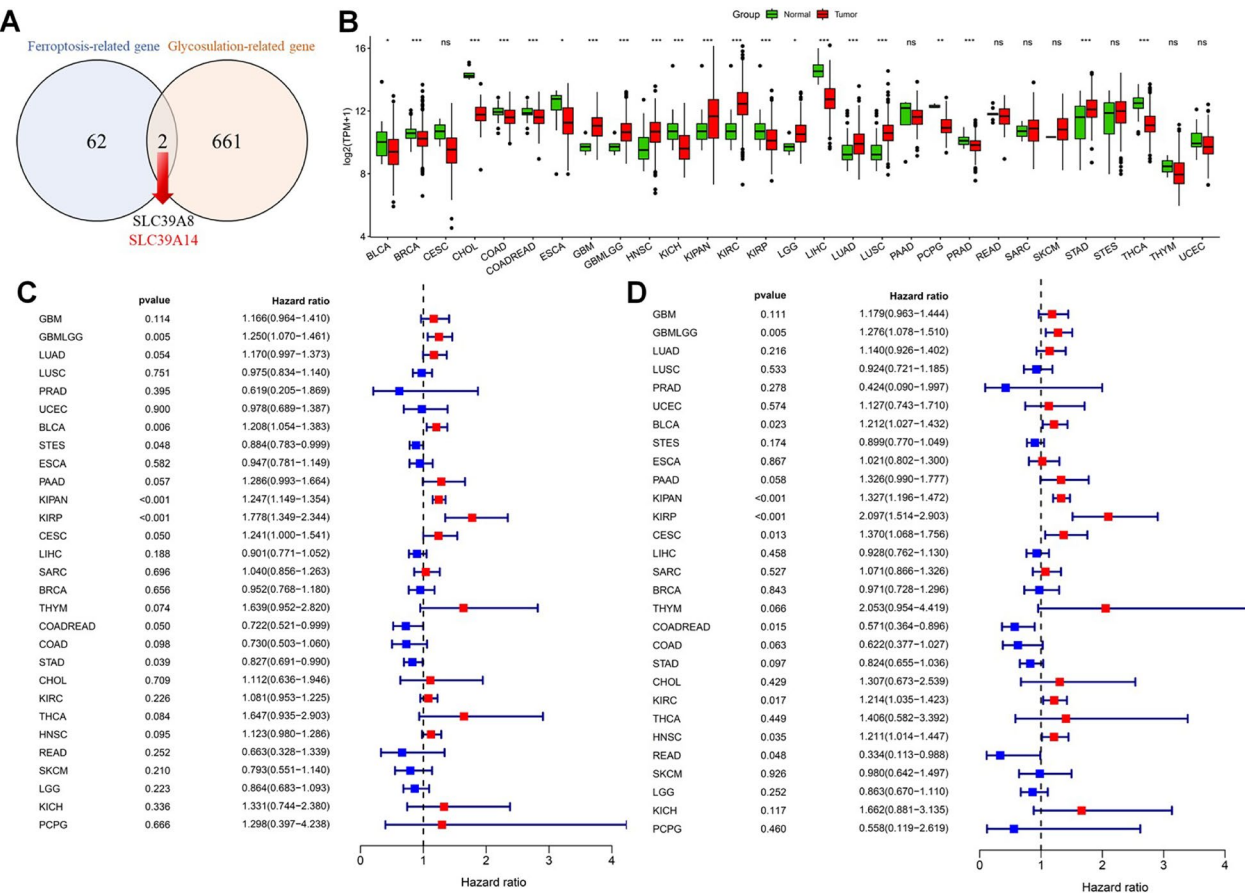


Fig. 6 Identification and pan-cancer validation of SLC39A14 as a candidate gene linking ferroptosis and glycosylation pathways. **(A)** Venn diagram showing the overlap between ferroptosis-related genes (blue, 64 genes) and glycosylation-related genes (orange, 663 genes), identifying SLC39A8 and SLC39A14 as shared candidates. **(B)** Differential expression analysis of SLC39A14 across cancer types in TCGA, comparing tumor tissues (red) with normal tissues (green). Statistical significance is indicated as $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), and ns, not significant. Forest plots of univariate Cox regression analysis showing associations of SLC39A14 expression with **(C)** OS and **(D)** DSS across cancer types. Red squares indicate hazard ratios (HRs) of > 1 (associated with poor prognoses), and blue squares indicate HRs of < 1 (associated with favorable prognoses). Horizontal lines represent 95% confidence intervals

aggressive, whereas low-score tumors may maintain homeostatic or differentiated functions.

To assess how SLC39A14 impacts TME, we analyzed its correlations with ESTIMATE-derived immune and stromal scores. In KIRC, SLC39A14 expression was correlated with both immune and stromal scores ($p < 2.2 \times 10^{-16}$). Similar trends were observed in GBM, and LGGs (Fig. 8A-D). These findings imply that SLC39A14 may shape both immune and ECM dynamics, possibly promoting a fibrotic or immunosuppressive microenvironment that enhances tumor resilience.

Single-cell transcriptomic analysis delineates slc39a14-driven ferroptosis resistance and immune-stromal reprogramming in kirc and gbm

To validate the multi-omics findings, we performed a single-cell RNA-Seq analysis for KIRC (GSE159115) and GBM (GSE182109), two cancer types identified by our pan-cancer multi-omics analysis as having SLC39A14

prognostic associations. In KIRC (Supplementary Figure S3A, Fig. 9A-E), single-cell profiling revealed that SLC39A14 expression was markedly enriched in tumor-associated macrophages (TAMs), natural killer (NK) cells, and fibroblast-like stromal populations rather than epithelial tumor cells alone. These SLC39A14-high immune-stromal clusters co-expressed markers of immunosuppressive immune cells, indicating an immune-excluded microenvironment. This localization aligns with the thrombin signaling and cytoskeletal remodeling pathways enriched in high-SLC39A14 KIRC tumors. Mechanistically, manganese import via SLC39A14 likely fuels glycosyltransferase activity within these stromal cells, altering the glycosylation of cytokine receptors and ECM proteins, thereby reinforcing immune evasion and ferroptosis resistance [43]. In GBM (Supplementary Figure S3B, Fig. 10A-G), the single-cell analysis showed a distinct pattern: SLC39A14 was predominantly expressed in T cells, mesenchymal-like glioma cells,

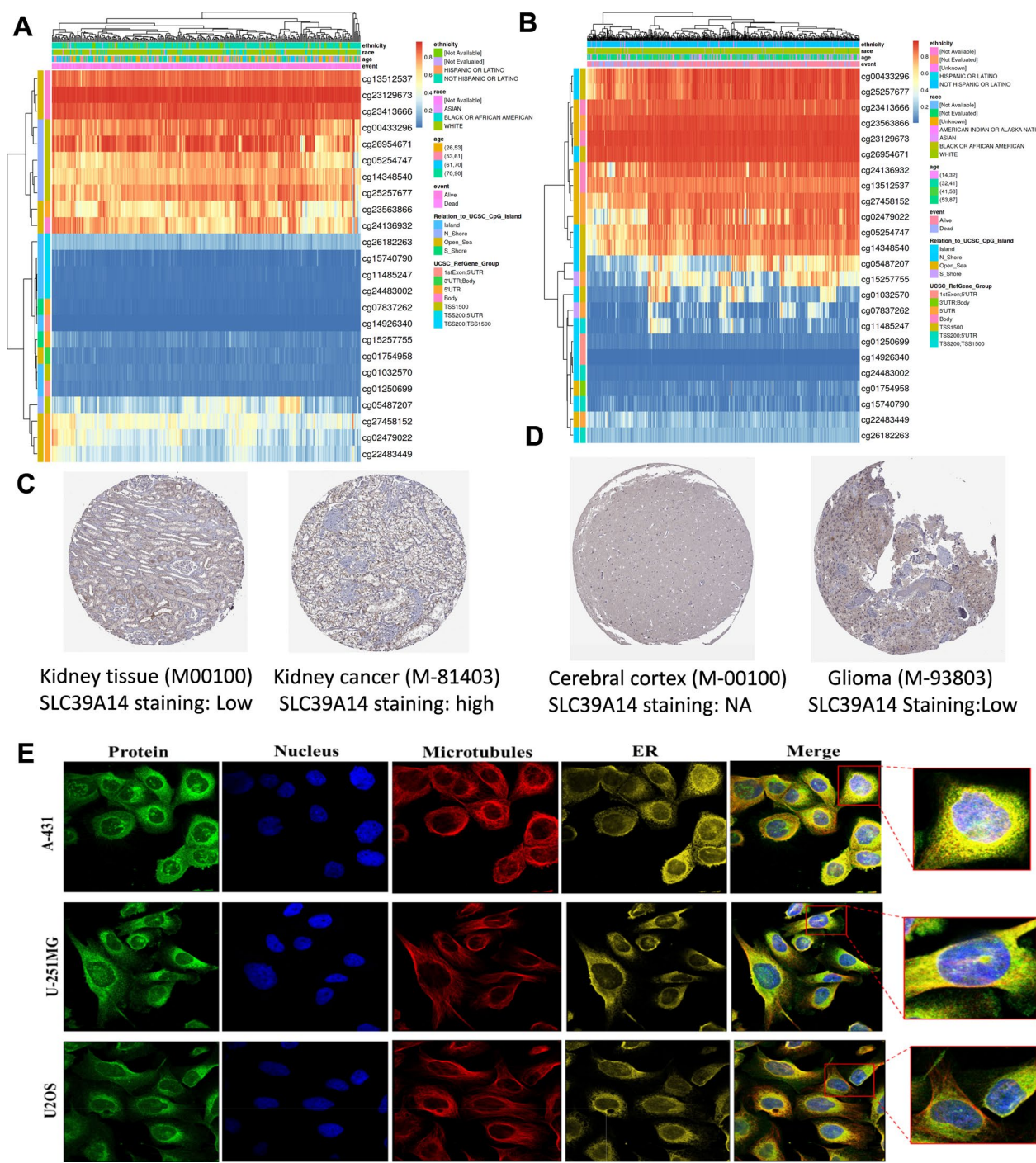


Fig. 7 DNA methylation profiles and protein-level validation of SLC39A14's expression in cancer. Heatmaps of DNA methylation levels for SLC39A14-associated CpG sites in **(A)** KIRC and **(B)** LGGs samples from TCGA. Rows correspond to CpG sites and columns to individual tumor samples. Clinical annotations (ethnicity, age, gender, and CpG island relation) are shown along the axes. Immunohistochemistry (IHC) from the Human Protein Atlas (HPA) showing SLC39A14 protein expression in **(C)** normal kidney versus kidney tumor tissues and **(D)** glioma tissues. Tumors display positive staining intensities and higher positivity compared to normal tissues. **(E)** Immunofluorescence microscopy of SLC39A14 in human cancer cell lines: A-431 (squamous cell carcinoma), U-251MG (glioblastoma), and U2OS (osteosarcoma). SLC39A14 is visualized in green, nuclei (DAPI) in blue, and cytoskeletal components in red. Merged images (yellow) indicate co-localization, with magnified insets showing subcellular localization

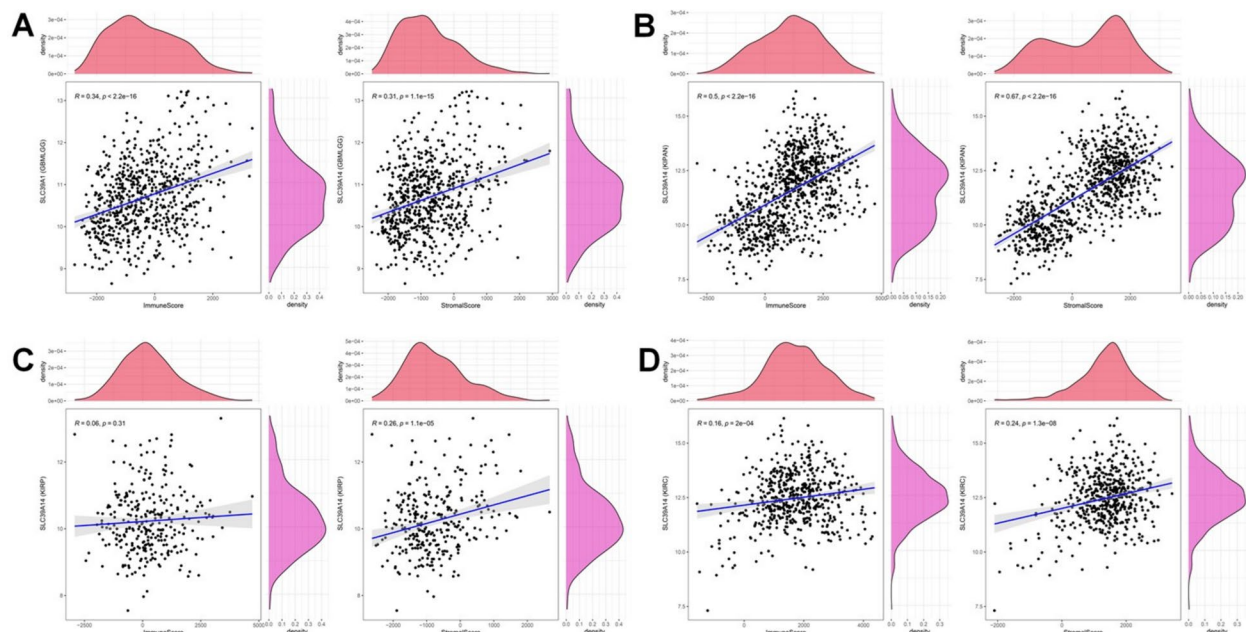


Fig. 8 Association of SLC39A14 expression with immune and stromal cell infiltration across cancer types. Pearson correlation plots showing relationships between SLC39A14 expression and immune scores (left) or stromal scores (right) derived from ESTIMATE analysis, reflecting immune and stromal cell infiltration in the tumor microenvironment. Representative cancer types include **(A)** GBMLGGs, **(B)** KIPAN, **(C)** KIRP, and **(D)** KIRC. Shaded areas indicate 95% confidence intervals for the regression line. R denotes Pearson's correlation coefficients; p values indicate statistical significance. The positive correlations were observed in GBMLGG and GBM, suggesting an immunomodulatory role for SLC39A14

and hypoxic perivascular niches. These cell clusters displayed upregulation of angiogenesis mediators and oxidative stress-adapted state. Furthermore, the single-cell insights support our study's rationale that SLC39A14 is not a uniform epithelial oncogene but a microenvironmental modulator that couples ferroptosis suppression with aberrant glycosylation and immune remodeling. By enabling stromal fibrosis, immune evasion, and redox resilience, SLC39A14 establishes a ferroptosis-resistant ecosystem that drives poor clinical outcomes in both KIRC and GBM. Therapeutically, this suggests that targeting SLC39A14 or its manganese-dependent enzymatic network could sensitize these tumors to ferroptosis inducers and immune checkpoint blockade, providing a translationally relevant strategy for overcoming therapeutic resistance [44, 45].

Pathway enrichment analysis of SLC39A14 associated degs in kirc and gbm

To elucidate the downstream molecular mechanisms driven by SLC39A14, we performed pathway enrichment analyses using the MetaCore platform, focusing on DEGs between high- and low-SLC39A14 expression groups. Results were analyzed separately for KIRC and GBM, highlighting both shared and context-specific signaling pathways. In KIRC, MetaCore enrichment analysis revealed three major oncogenic pathways closely tied to SLC39A14 biology (Fig. 11). Previous studies

demonstrated that tissue factor (TF) activation leads to protease-activated receptor (PAR)-mediated signaling, promoting cancer cell proliferation and invasion. Our findings further indicated that TF signaling interacts with VEGF pathways, facilitating endothelial cell migration and vascular remodeling, underscoring the potential of targeting TF signaling to inhibit tumor growth and metastasis [46, 47]. The DNA damage intra S phase checkpoint pathway (Supplementary Figure S4) was significantly enriched, with co-expressed genes including CHEK1, RAD51, ATR, and MCM family members [48]. This enrichment suggests that SLC39A14 is associated with heightened DNA replication stress and impaired checkpoint control, conditions that promote genomic instability and tumor progression. Biologically, this is consistent with our findings that SLC39A14 supports cell survival under oxidative and metabolic stress, potentially by buffering replication stress and facilitating DNA repair programs [49]. The Hippo/YAP signaling pathway (Supplementary Figure S5) was prominently activated, with reduced upstream kinase activity and enhanced YAP/TAZ signaling. This shift supports tumor cell proliferation and survival, consistent with our observation that SLC39A14 promotes ferroptosis resistance and glycosylation-driven remodeling, both of which converge on YAP/TAZ activation to maintain oncogenic signaling [50, 51]. The enrichment of actin cytoskeleton remodeling through Rho GTPase effectors (Supplementary Figure

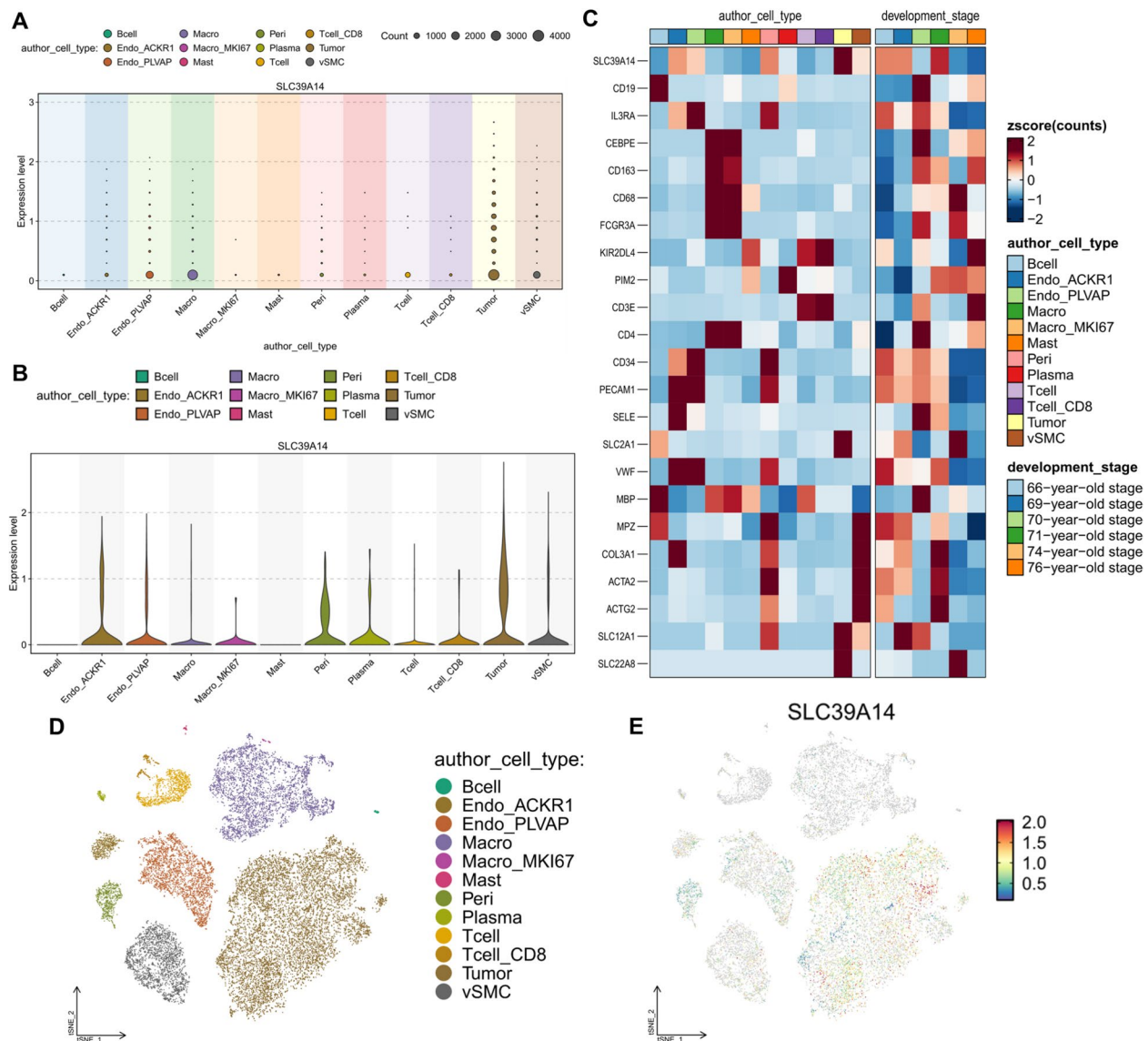


Fig. 9 Single-cell transcriptomic profiling of SLC39A14 expression in KIRC (GSE159115)

S6) highlights a mechanism for enhanced motility and epithelial mesenchymal transition (EMT). Biologically, this aligns with our finding that SLC39A14 fosters vascular invasion in renal carcinomas by reshaping cytoskeletal dynamics and extracellular matrix organization. Another pathway which is the GDNF–RET receptor pathway (Supplementary Figure S7) was enriched, implicating sustained activation of PI3K–AKT and MAPK cascades. This supports the role of SLC39A14 in reinforcing pro-survival signaling while modulating redox balance via manganese-dependent glycosylation. Together, these MetaCore findings in KIRC suggest that SLC39A14 drives proliferation, invasion, and ferroptosis evasion in KIRC by rewiring YAP signaling, cytoskeletal plasticity, and RET-mediated growth program [52].

In GBM, enrichment results are visualized in Fig. 12, with largely overlapping but distinct pathway profiles. SLC39A14 associated DEGs were enriched in VEGF-driven angiogenesis and ER stress response pathways [53]. These enrichment pathways reveals that integrate SLC39A14's roles in ferroptosis resistance, stromal remodeling, and immune evasion. The systemic sclerosis endothelial/immune pathway (Supplementary Figure S8) was enriched, reflecting aberrant endothelial immune crosstalk. This again supports our findings that SLC39A14 is expressed in hypoxic perivascular niches, where manganese-driven glycosylation remodeling promotes fibrosis-like ECM deposition and vascular remodeling, establishing immune-excluded tumor zones [44, 54]. Further pathway, which is the MHC class I cross-presentation pathway (Supplementary Figure S9) was

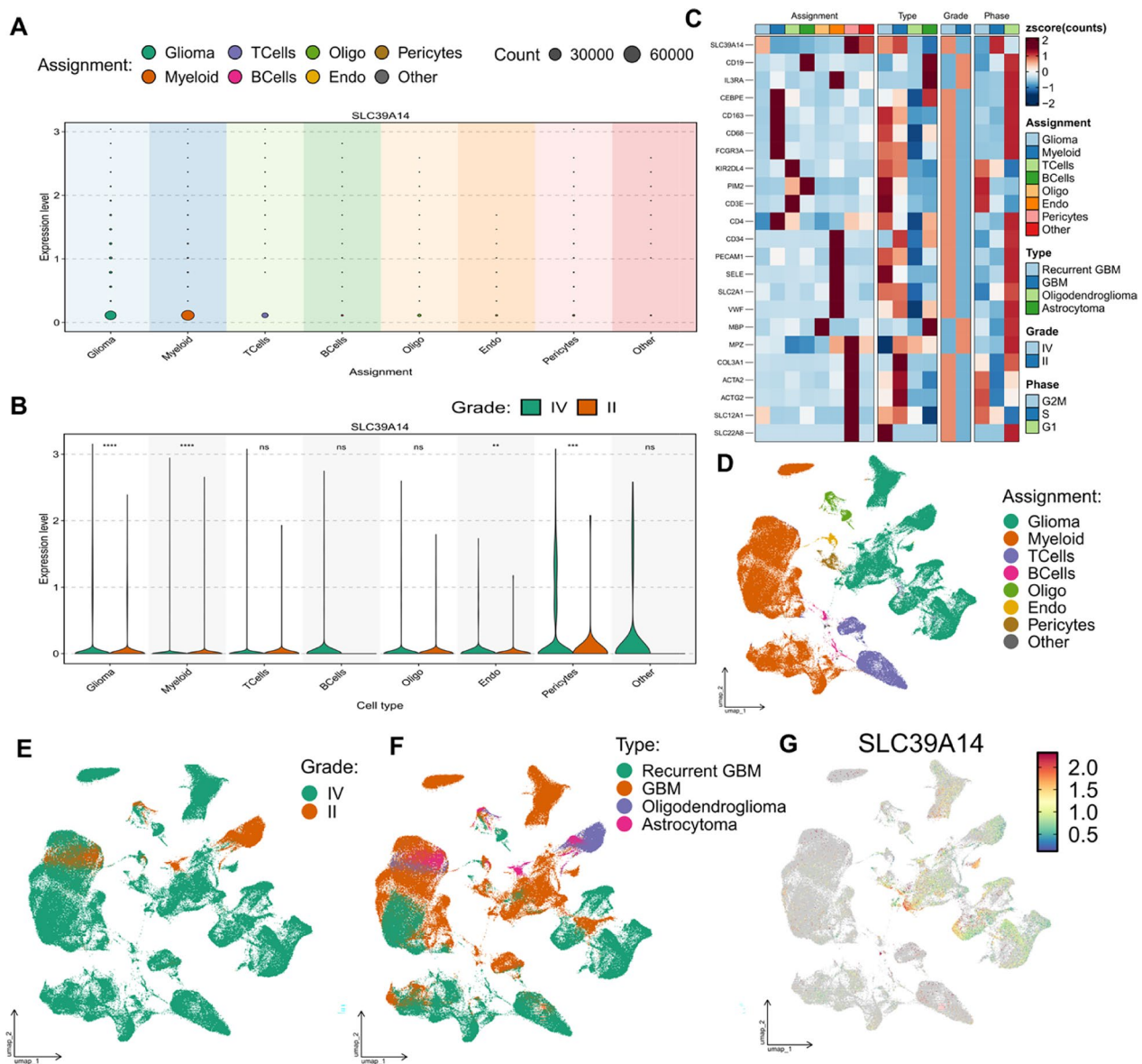


Fig. 10 Single-cell transcriptomic profiling of SLC39A14 expression in GBM (GSE182109). **(A)** Dot plot showing SLC39A14 expression levels across annotated cell types, with the dot size indicating cell proportions and the color intensity reflecting expression magnitudes. **(B)** Violin plots displaying SLC39A14 expression across cell types, stratified by GBM grade (II vs. IV), with statistical significance indicated. **(C)** Heatmap comparing SLC39A14 expression with canonical cell-type marker genes across cell types, GBM types, tumor grades, and cell-cycle phases. **(D)** UMAP visualization of annotated cell clusters by cell type. **(E)** UMAP plot showing the distribution of GBM grades (II vs. IV). **(F)** UMAP plot highlighting major GBM histological subtypes. **(G)** Feature plot showing transcript expression patterns of SLC39A14 on the UMAP projection, with the color scale indicating relative expression intensities

also enriched, suggesting impaired tumor antigen display. This aligns with our data showing that SLC39A14 modifies glycosylation of immune receptors and cytokine pathways, thereby dampening effective CD8⁺ T-cell recognition. Such alterations explain why high SLC39A14 tumors adopt an immune-evasive phenotype [55]. The macrophage and dendritic cell phenotype shift in cancer pathway (Supplementary Figure S10) indicated skewing toward immunosuppressive myeloid states, and resistance to therapy by maintaining ROS tolerance and stromal-immune remodeling [55] (Fig. 13).

Discussion

This study provides comprehensive insights into the dual role of SLC39A14 in ferroptosis and glycosylation, emphasizing its critical contributions to tumor biology and its potential as both a prognostic biomarker and therapeutic target. By integrating a multi-omics bioinformatics analysis across pan-cancer datasets, we highlighted how SLC39A14 links two fundamental pathways, ferroptosis, a form of iron-dependent cell death, and glycosylation, a pivotal post-translational modification, thus underscoring its multifaceted influence on tumor

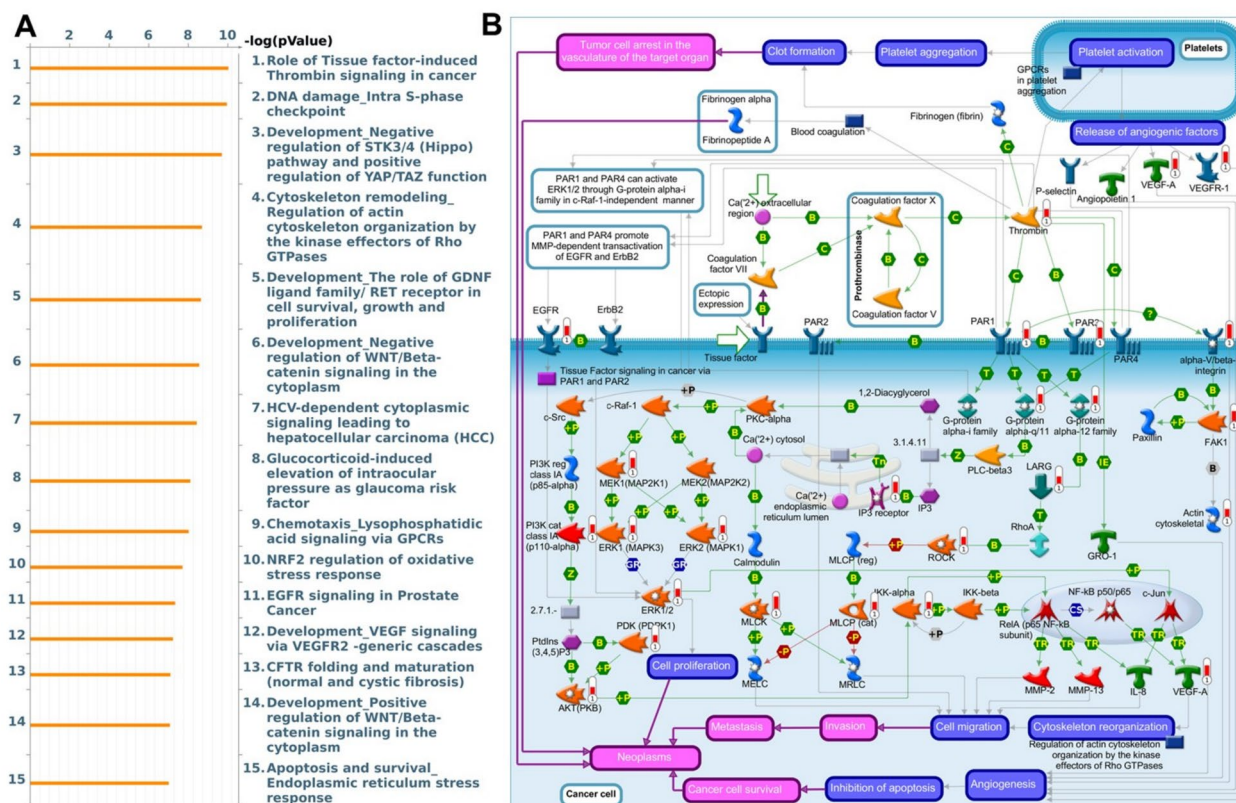


Fig. 11 MetaCore pathway enrichment analysis revealing key signaling pathways associated with SLC39A14 in KIRC progression. **(A)** Bar chart ranking significantly enriched pathways by $-\log(p)$ values, highlighting the prominent involvement of thrombin signaling, the DNA damage response, Hippo/YAP signaling, cytoskeletal remodeling, WNT/ β -catenin regulation, the NRF2 oxidative stress response, VEGF signaling, and endoplasmic reticular stress response. **(B)** Schematic representation of the top-ranked pathway, “Tissue factor (TF)-induced thrombin signaling in cancer”, showing molecular interactions that regulate tumor cell proliferation, survival, cytoskeletal reorganization, angiogenesis, migration, and metastasis. Key nodes and signaling modules are indicated, illustrating potential mechanisms by which SLC39A14 may contribute to tumor progression and highlighting putative therapeutic intervention points

progression and the TME [56–58]. The identification of SLC39A14 from the overlap between glycosylation- and ferroptosis-related gene sets positions it as a unique dual-functional regulator. Functionally, SLC39A14 may modulate iron influx, ROS buffering, and protein glycosylation patterns, all of which could synergistically promote ferroptosis resistance and oncogenesis. These findings are consistent with prior reports linking SLC39A14 to manganese/zinc transport, NRF2 activation, and inflammatory signaling cascades [54]. The multivariate Cox analysis revealed that SLC39A14 significantly contributes to OS across diverse cancer types. These findings suggest that ferroptosis-related genes, particularly SLC39A14, modulate oxidative stress, lipid peroxidation, and iron homeostasis, thereby influencing tumor aggressiveness. Future mechanistic studies are warranted to explore its precise function in regulating ferroptotic vulnerability in cancer cells. The stratification of ferroptosis scores across tumor types revealed marked inter-cancer variability in both expression and clinical relevance. High ferroptosis scores largely driven by expressions of key regulators such

as SLC39A14 were significantly associated with poor outcomes in tumors such as gliomas, and renal carcinomas. Kaplan–Meier survival analyses confirmed that these scores serve as reliable predictors of clinical prognosis. Interestingly, the impact of SLC39A14 varied across tumor types, reflecting the context-specific interactions between ferroptosis and the TME, including immune, stromal, and metabolic components. These insights emphasize the necessity of developing tumor-specific therapeutic strategies that target SLC39A14 or ferroptosis-related vulnerabilities in a personalized manner [59]. Importantly, our data suggest that SLC39A14-driven ferroptosis suppression and glycosylation remodeling are metabolically coupled to glutamine metabolism, generating α -ketoglutarate and other intermediates that support mitochondrial oxidative phosphorylation and redox balance. At the same time, glutamine-derived carbon fuels the hexosamine biosynthetic pathway, providing uridine diphosphate N-acetylglucosamine (UDP-GlcNAc) for N- and O-linked glycosylation [60]. In SLC39A14-high tumors, Mn^{2+} import enhances glycosyltransferase

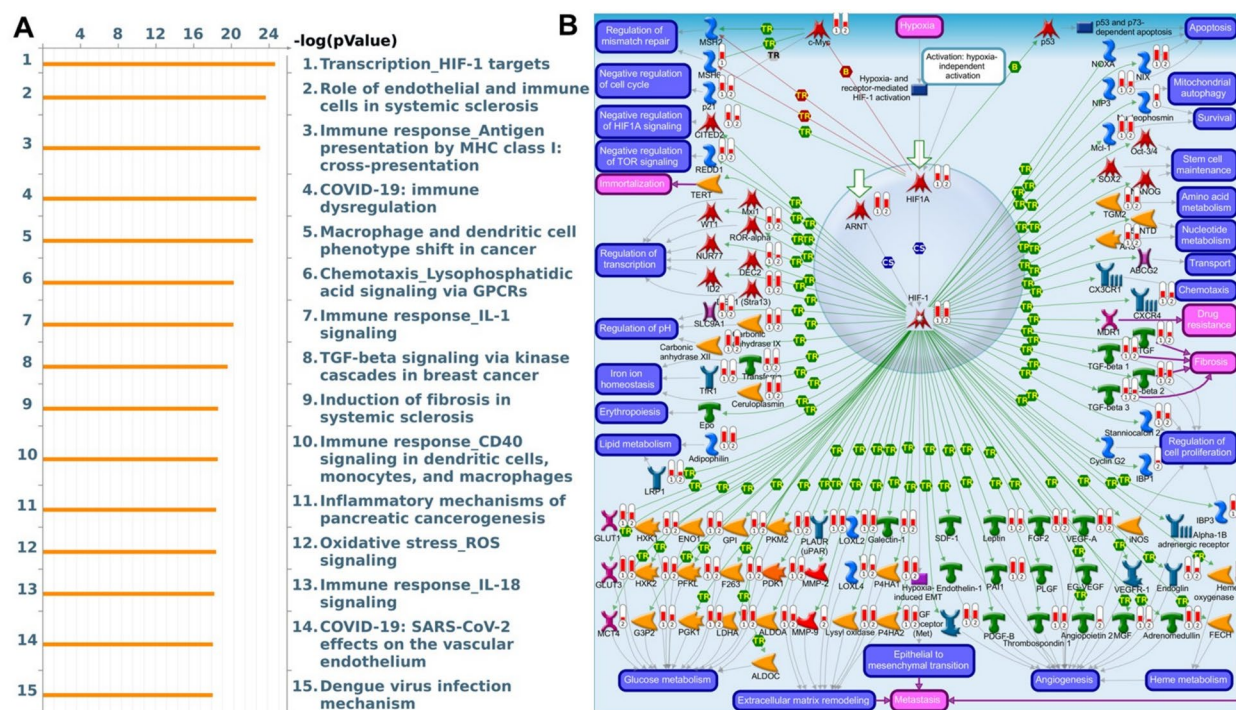


Fig. 12 MetaCore pathway enrichment analysis revealing key signaling pathways associated with SLC39A14 in GBM. **(A)** Bar chart ranking significantly enriched pathways by $-\log(p)$ values, highlighting major signaling networks implicated in tumor biology. **(B)** Representative schematic of the top-ranked pathway, "Transcription_HIF-1 targets", detailing molecular interactions and regulatory nodes involved in hypoxia-driven processes that promote brain cancer progression. Key effectors associated with cell proliferation, survival, angiogenesis, and metabolic adaptation are indicated. This analysis provides mechanistic insights into how SLC39A14 may contribute to tumor development and identifies potential therapeutic targets for interventions

activity, while Fe^{2+} import increases the ferroptotic potential that is counteracted by the metabolism-related antioxidant capacity such as NADPH and glutathione (GSH) synthesis [61]. This metabolic configuration ensures continuous glycosylation of ECM components, receptors, and immune checkpoints, while sustaining glutathione peroxidase 4 (GPX4)-mediated lipid peroxide detoxification, thereby directly linking mitochondrial anaplerosis to ferroptosis resistance and immune evasion. Interestingly, the threshold for the clinical impact of SLC39A14 expression and ferroptosis scores varied considerably across different cancer types, raising important questions about the biological and environmental factors influencing these discrepancies. One possible explanation lies in population-level heterogeneity in baseline metabolic health, genetic predispositions, dietary habits (such as fat- or carbohydrate-rich diets), age distributions, and epigenetic regulation patterns [62–65]. For instance, cancers such as GBM and KIRC exhibited pronounced associations with high SLC39A14 expression and ferroptosis resistance, which may reflect inherent differences in cellular redox states, the iron metabolism capacity, or the manganese-dependent glycosylation machinery driven by both host factors (such as age-related antioxidant decline) and tumor-intrinsic features (such as hypoxia or NRF2 activation) [66–68].

Additionally, the epigenetic landscape, as evidenced by promoter hypomethylation patterns, could be differentially shaped by external exposures such as inflammation, microbiota-derived metabolites, or long-term dietary patterns, further contributing to inter-tumoral variability [68, 69]. These threshold effects underscore the need for hypothesis-generating, stratified studies that integrate host metabolic profiling with tumor omics to clarify why certain tumors exhibit higher dependency on ferroptosis evasion mechanisms. Future studies should therefore incorporate population-specific data especially involving age, dietary manganese intake, and glyco-redox profiles to refine predictive thresholds and optimize personalized targeting of SLC39A14 [70, 71].

The cancer-type-specific survival analyses further validated the prognostic utility of SLC39A14. In KIRC and GBM, high SLC39A14 expression was significantly associated with poor survival. These correlations suggest that elevated SLC39A14 expression may facilitate ferroptosis resistance, enabling tumor cells to withstand oxidative damage and adopt invasive, therapy-resistant phenotypes [56, 72]. Additionally, its overexpression in these cancer types might be linked to enhanced metabolic flexibility and immune evasion, both key traits of tumor aggressiveness [73]. The significant correlation between SLC39A14 expression and immune/stromal scores suggests a role in

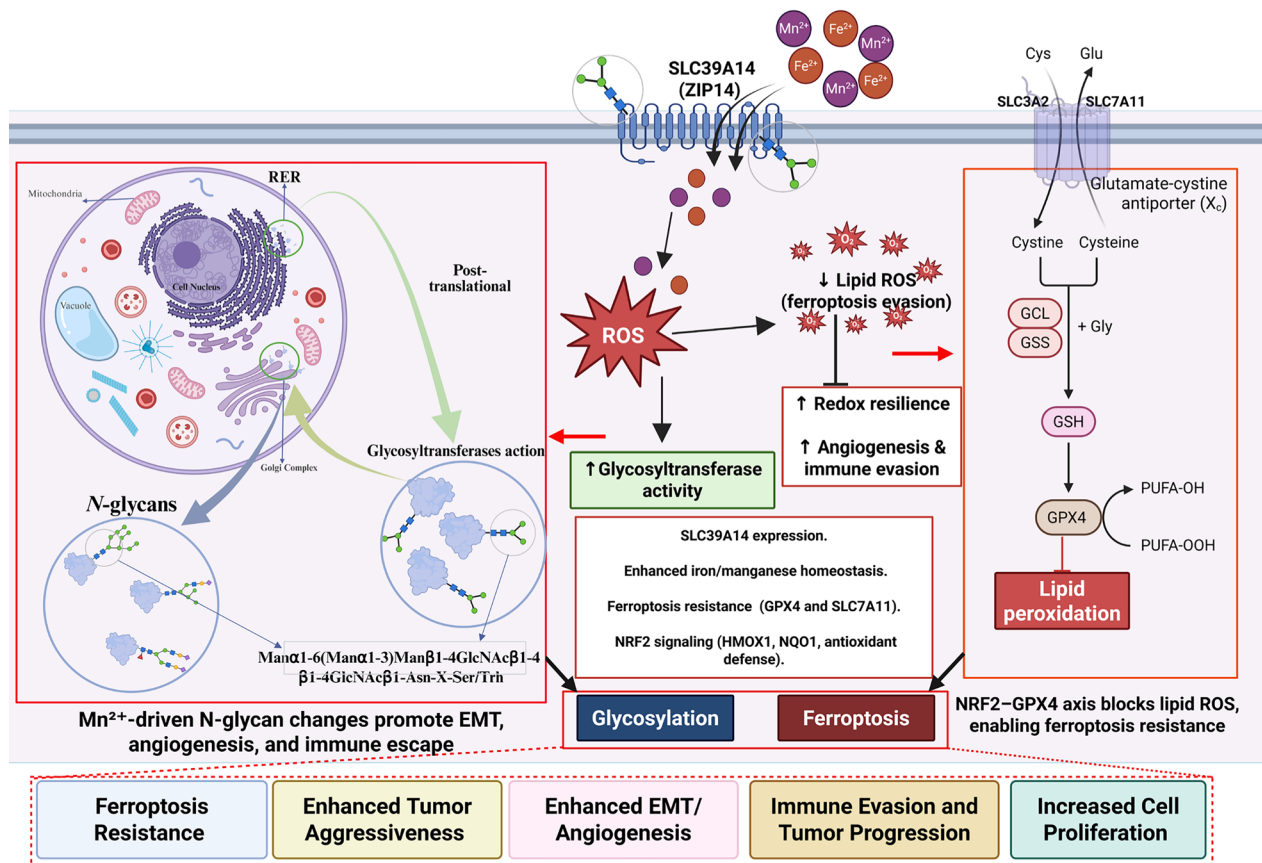


Fig. 13 Crosstalk of SLC39A14 with glycosylation and ferroptosis resistance in tumor progression. Overexpression of SLC39A14 promotes Mn^{2+} and Fe^{2+} uptake, enhancing Golgi glycosyltransferase activity and N-glycosylation input. These glycosylation changes drive tumor aggressiveness, the epithelial-to-mesenchymal transition (EMT), angiogenesis, and immune evasion. Concurrently, glutamine metabolism supports glycolysis, sustaining redox homeostasis. The cystine/glutamate antiporter (X_c^-) imports cystine for glutathione (GSH) synthesis, enabling GPX4-mediated inhibition of lipid peroxidation and ferroptosis. Together, these pathways promote ferroptosis resistance, redox resilience, and tumor progression

shaping the immune landscape and fibrotic remodeling of tumors. Elevated SLC39A14 expression was positively associated with immune-excluded or immune-suppressed phenotypes, especially in KIRC and GBM. These findings raise the possibility that SLC39A14 may influence immune evasion via altered cytokine glycosylation, antigen presentation, or immune modulation [74]. These data suggest that SLC39A14 supports immune exclusion, oxidative stress adaptation, and stromal activation, reinforcing its role as a modulator of the TME and ferroptotic escape [75, 76]. The pathway enrichment analysis revealed significant involvement of tissue factor-induced thrombin signaling, DNA damage responses, cytoskeleton remodeling, and WNT/ β -catenin signaling in KIRC. Each of these pathways contributes to cancer pathophysiology, supporting their relevance as potential therapeutic targets. Tissue factor-induced thrombin signaling, the most significantly enriched pathway, plays a crucial role in tumor-associated thrombosis, angiogenesis, and metastasis. However, in GBM, enrichment of VEGF signaling, Hippo-YAP, and NRF2 oxidative stress defense

highlighted its role in maintaining hypoxia-adapted, immune-resistant niches and sustaining glioma stemness [77]. Notably, NRF2 target genes such as HMOX1, NQO1, and GCLC were upregulated in SLC39A14-high GBM, suggesting ferroptosis evasion as a key mechanism of survival under oxidative stress [78].

Despite the breadth of our integrative analysis, the study still has some limitations and requires functional validation, including SLC39A14 perturbation, ferroptosis sensitivity assays, glycoproteomics and spatial or multiplex imaging to define interactions with immune and stromal compartments [79]. Prospective, clinically annotated cohorts are needed to address TCGA heterogeneity, and mechanistic experiments should test whether SLC39A14 mediated manganese transport modulates glycosylation, glutamine metabolism, and redox regulation during cancer progression.

Conclusions

Collectively, this study identified SLC39A14 as a novel molecular hub that orchestrates critical oncogenic programs by integrating ferroptosis suppression, glycosylation remodeling, and immune modulation. Through a pan-cancer, multi-omics approach, we demonstrated that SLC39A14 is overexpressed in diverse malignancies, particularly in KIRC and GBM, and correlates with poor outcomes. Meanwhile, SLC39A14 modulates oxidative stress, immune infiltration, endothelial remodeling, and cell-cycle regulation. Single-cell transcriptomics further revealed its preferential expression in immune-stromal niches associated with ferroptosis resistance and immune escape. Its associations with TF-induced thrombin signaling, VEGF angiogenesis, Hippo-YAP, and NRF2 pathways highlight its central role in tumor progression. Together, these findings justify further investigation of SLC39A14 as both a clinical biomarker and precision oncology target, especially in therapy-resistant and immune-excluded tumors (Fig. 13).

Abbreviations

TF	Tissue factor
PAR	Protease-activated receptor
VEGF	Vascular endothelial growth factor
ER	Endoplasmic reticulum
CpG	Cytosine-phosphate-guanine
NK	Natural killer (cells)
EMT	Epithelial-to-mesenchymal transition
LASSO	Least Absolute Shrinkage and Selection Operator
TCGA	The Cancer Genome Atlas
OS	Overall survival
DSS	Disease-specific survival
PFI	Progression-free interval
DEG	Differentially expressed gene
GO	Gene ontology
KEGG	Kyoto Encyclopedia of Genes and Genomes
HPA	Human Protein Atlas
IF	Immunofluorescence
IHC	Immunohistochemistry
MSigDB	Molecular Signatures Database
ESTIMATE	Estimation of STromal and Immune cells in MAlignant Tumors using Expression data
scRNA-Seq	Single-cell RNA sequencing
t-SNE	T-distributed stochastic neighbor embedding
HR	Hazard ratio
CI	Confidence interval
FDR	False discovery rate
ROC	Receiver operating characteristic
DCA	Decision curve analysis
NRF2	Nuclear factor erythroid 2-related factor 2

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12935-025-04003-6>.

Additional file1

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Author contributions

Developed the concept and designed the study: Yi-Chun Chiang, Chih-Yang Wang, Sachin Kumar, Shun-Fa Yang, Yung-Kuo Lee. Performed data analysis and interpretation: Chung-Bao Hsieh, Kai-Fu Chang, Ching-Chung Ko, Chih-Hsuan Chang, Hui-Ru Lin, Chi-Jen Wu, Chien-Han Yuan, Do Thi Minh Xuan, Juan Lorell Ngadio, Dahlak Daniel Solomon, Fitria Sari Wulandari, Hung-Yun Lin. All authors have read and approved the final version of this manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval

Not applicable.

Consent for publication

All authors have read and agreed to the publication of the manuscript.

Competing interest

The authors declare no competing interests.

(A) Dot plot showing SLC39A14 expression levels across annotated cell types, with the dot size indicating cell proportions and the color intensity reflecting expression magnitudes. **(B)** Violin plots displaying the distribution of SLC39A14 expression across cell types. **(C)** Heatmap comparing SLC39A14 expression with canonical cell-type marker genes across multiple cell types and patient developmental stages. **(D)** t-SNE visualization of annotated cell clusters, colored by cell type. **(E)** t-SNE feature plot showing transcript expression patterns of SLC39A14, with the color scale indicating the relative expression intensity.

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